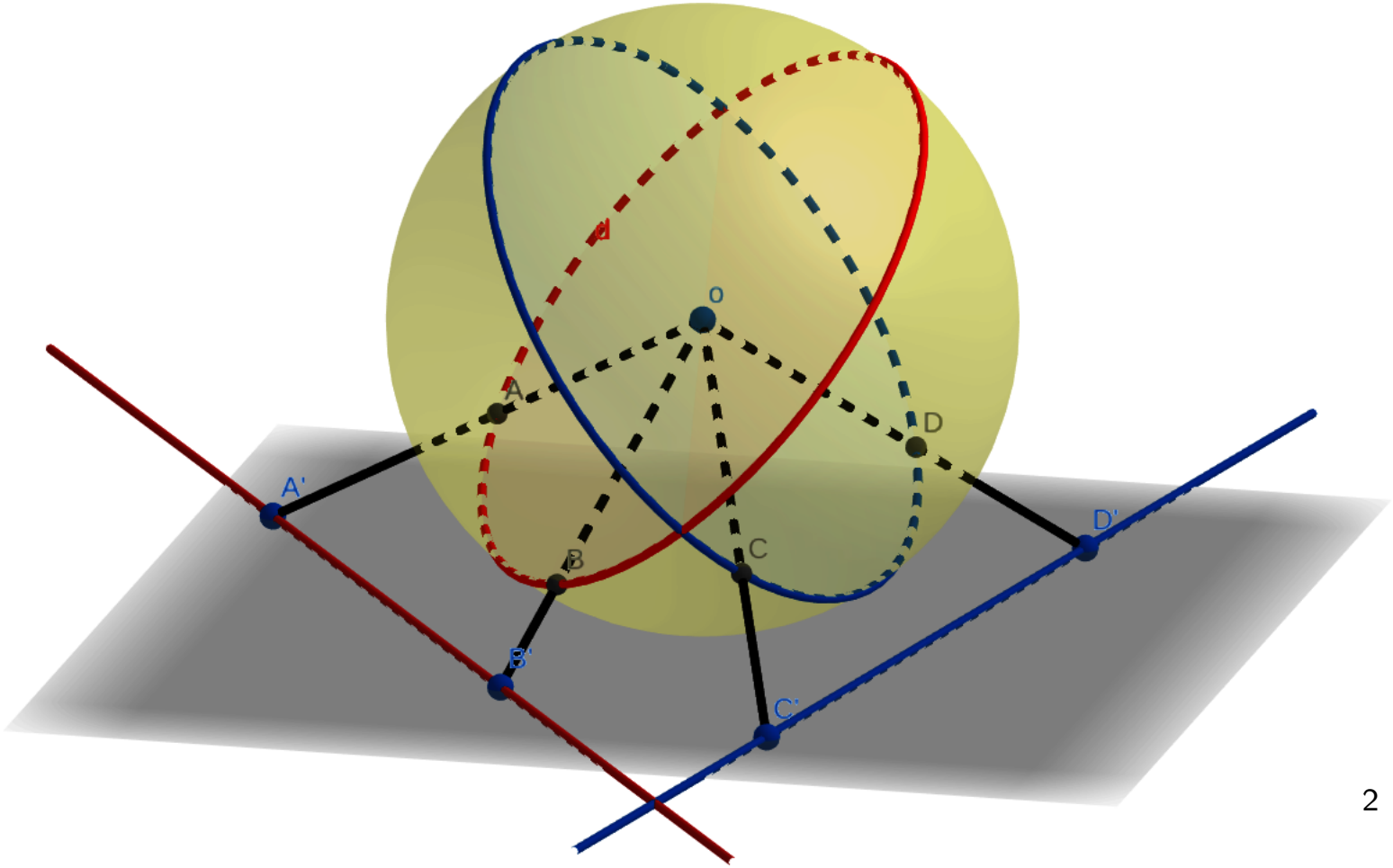
Gons and Holes in Projective Point Sets

Martin Balko, Manfred Scheucher, Pavel Valtr

Convex Sets in Projective Plane (Steinitz 1913)



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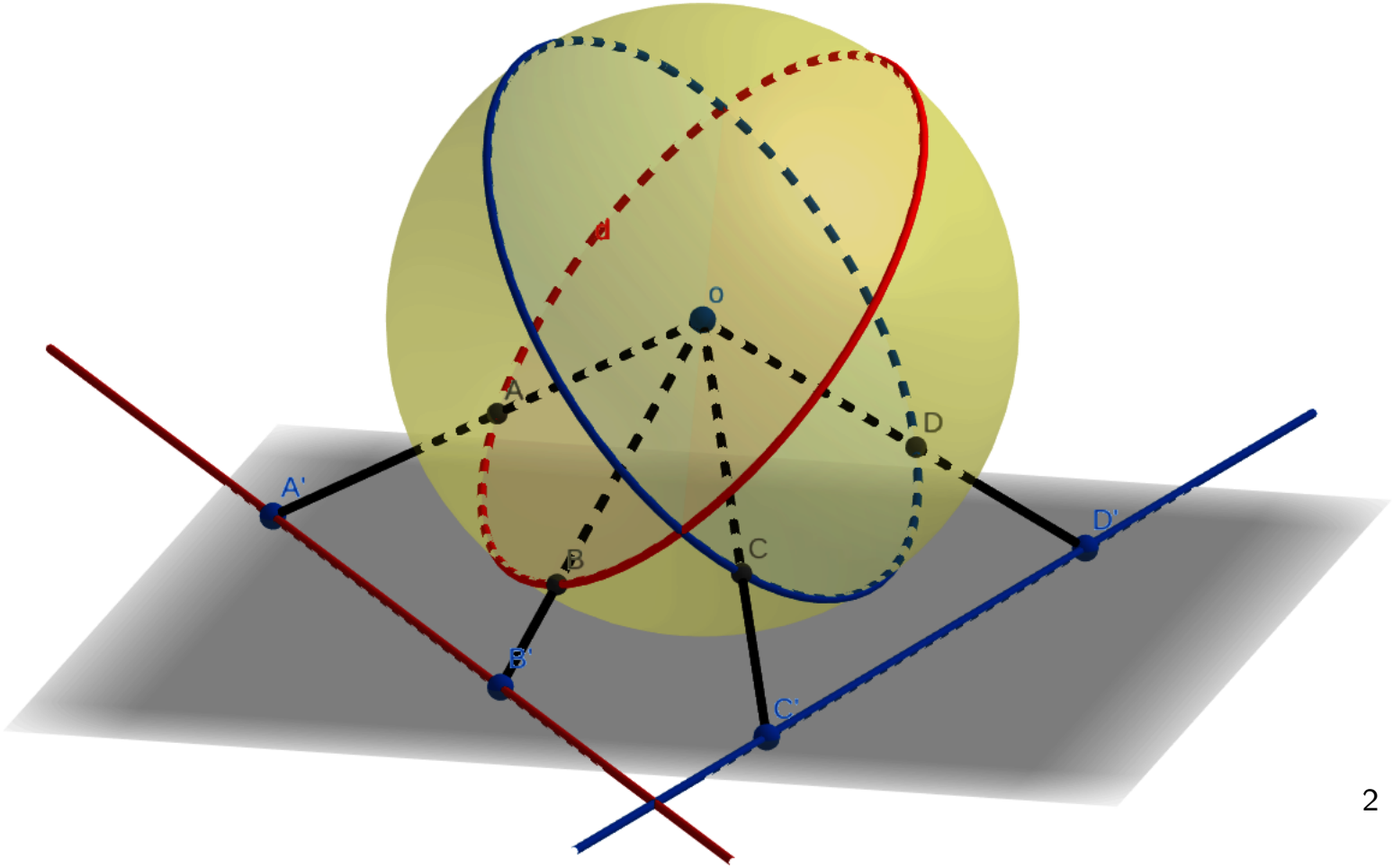
an affine line segment pq and (its complement) $\overline{pq} \setminus pq$



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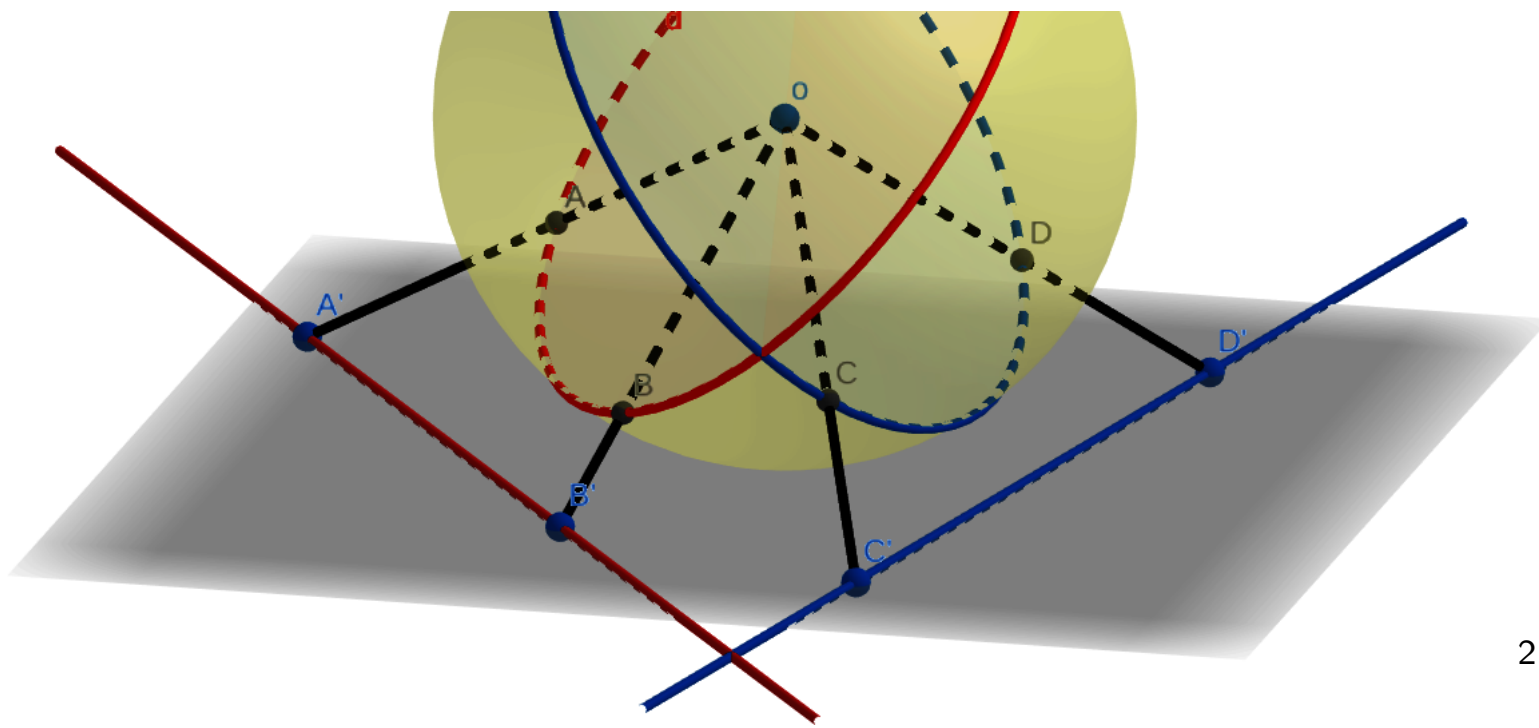

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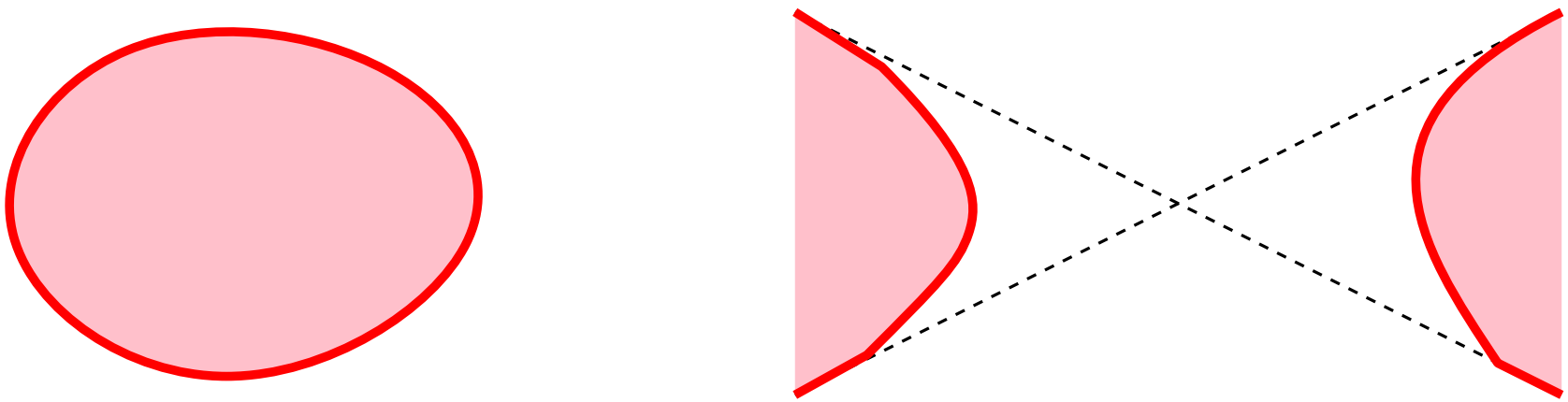

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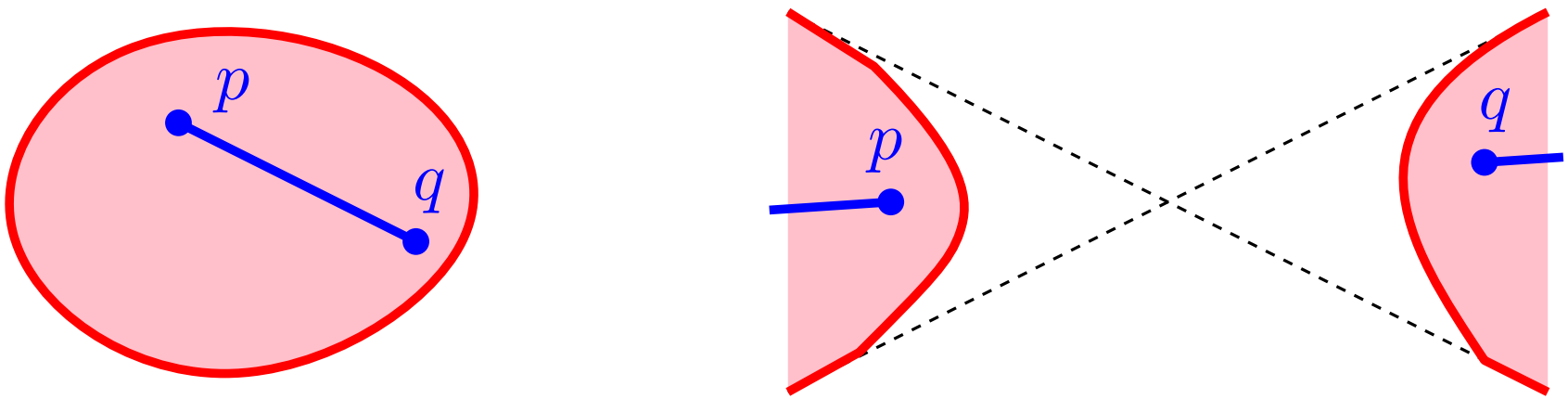

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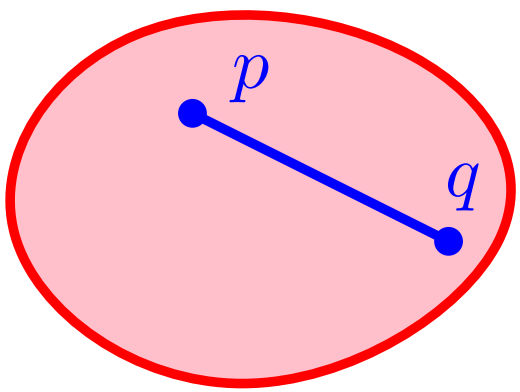
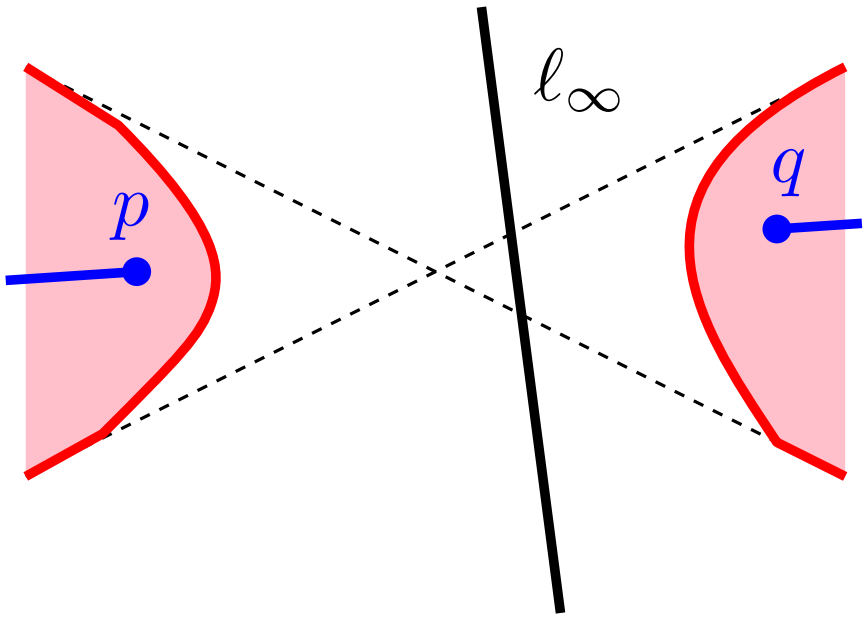
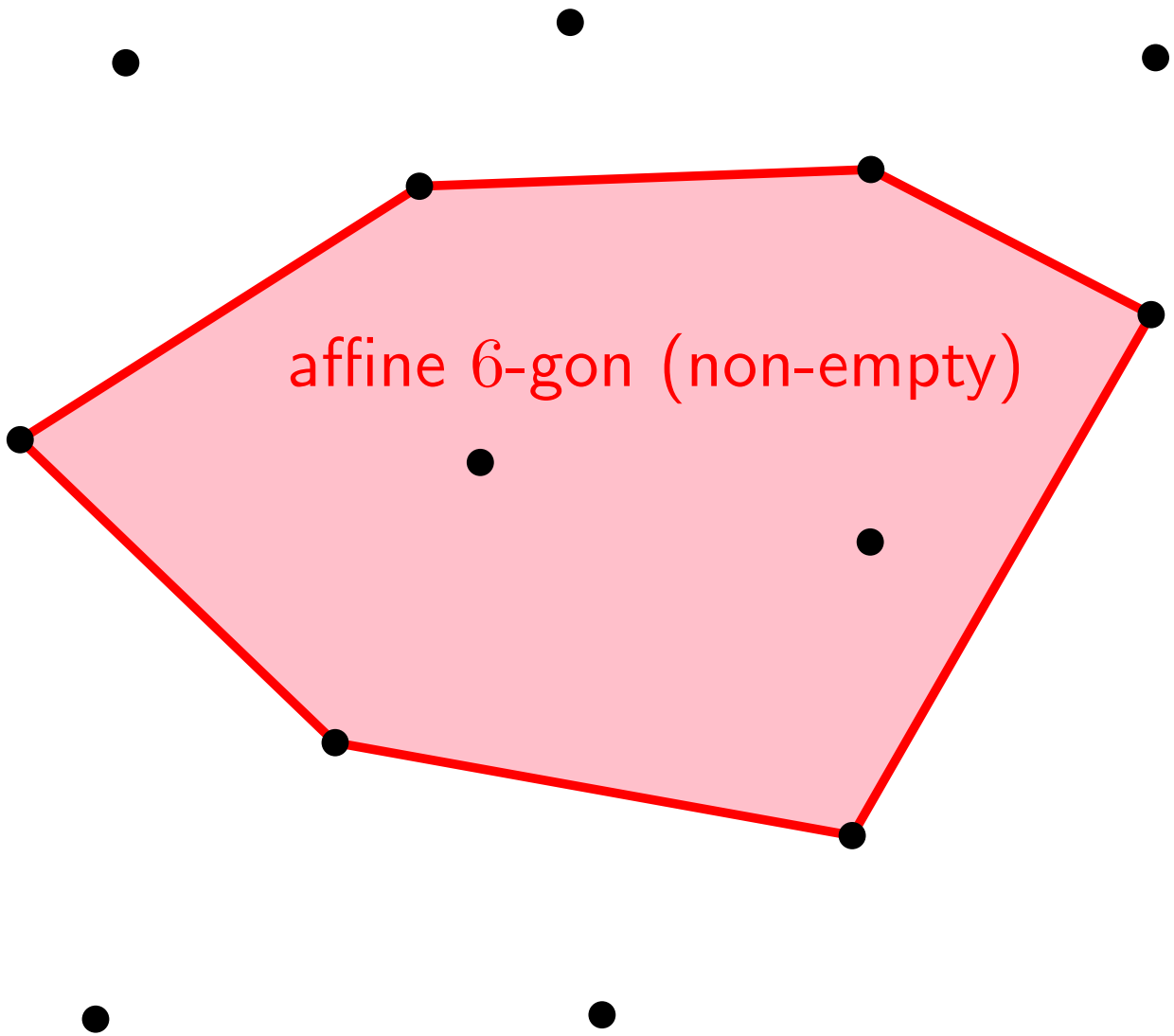




Projective Gons and Holes



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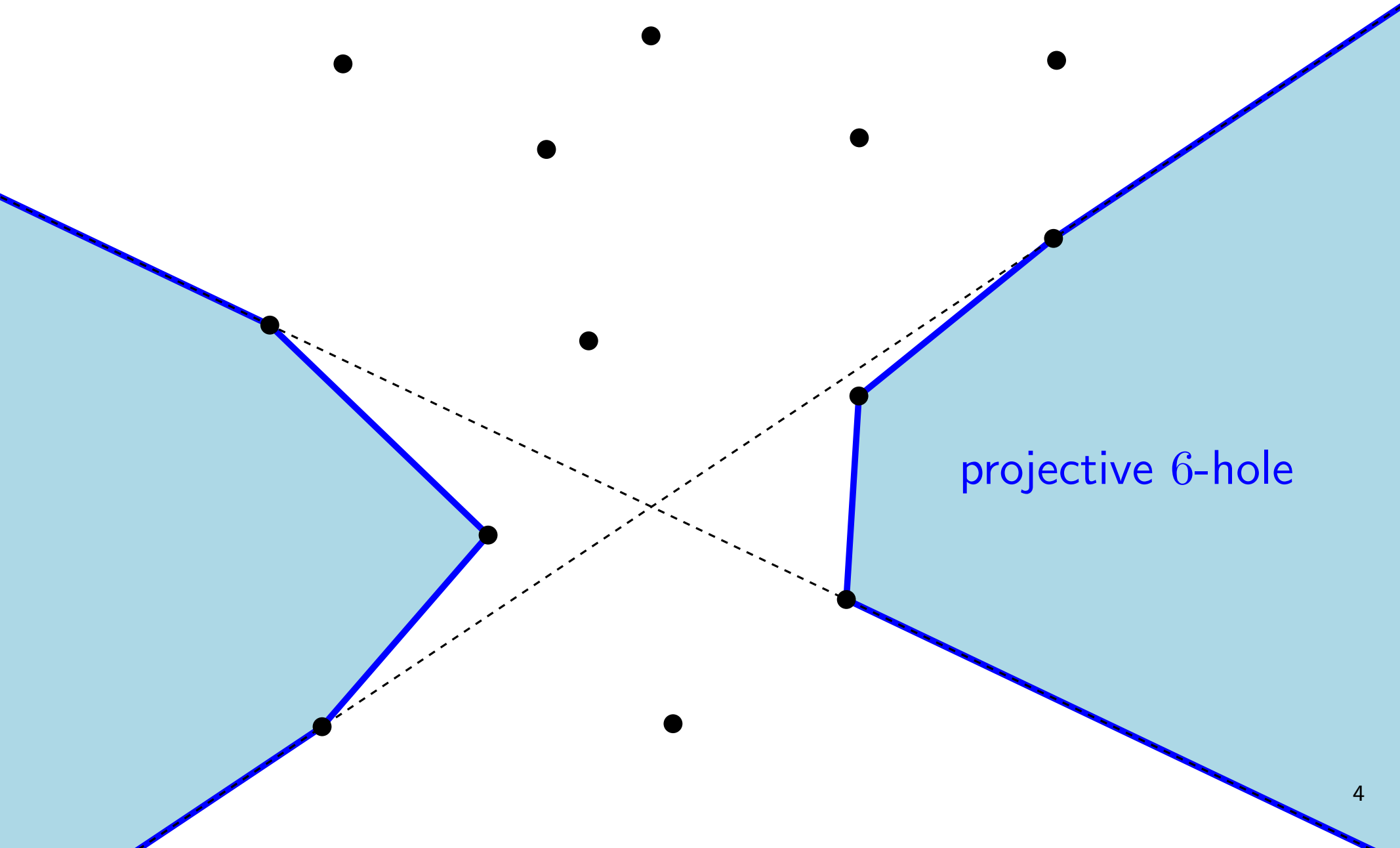

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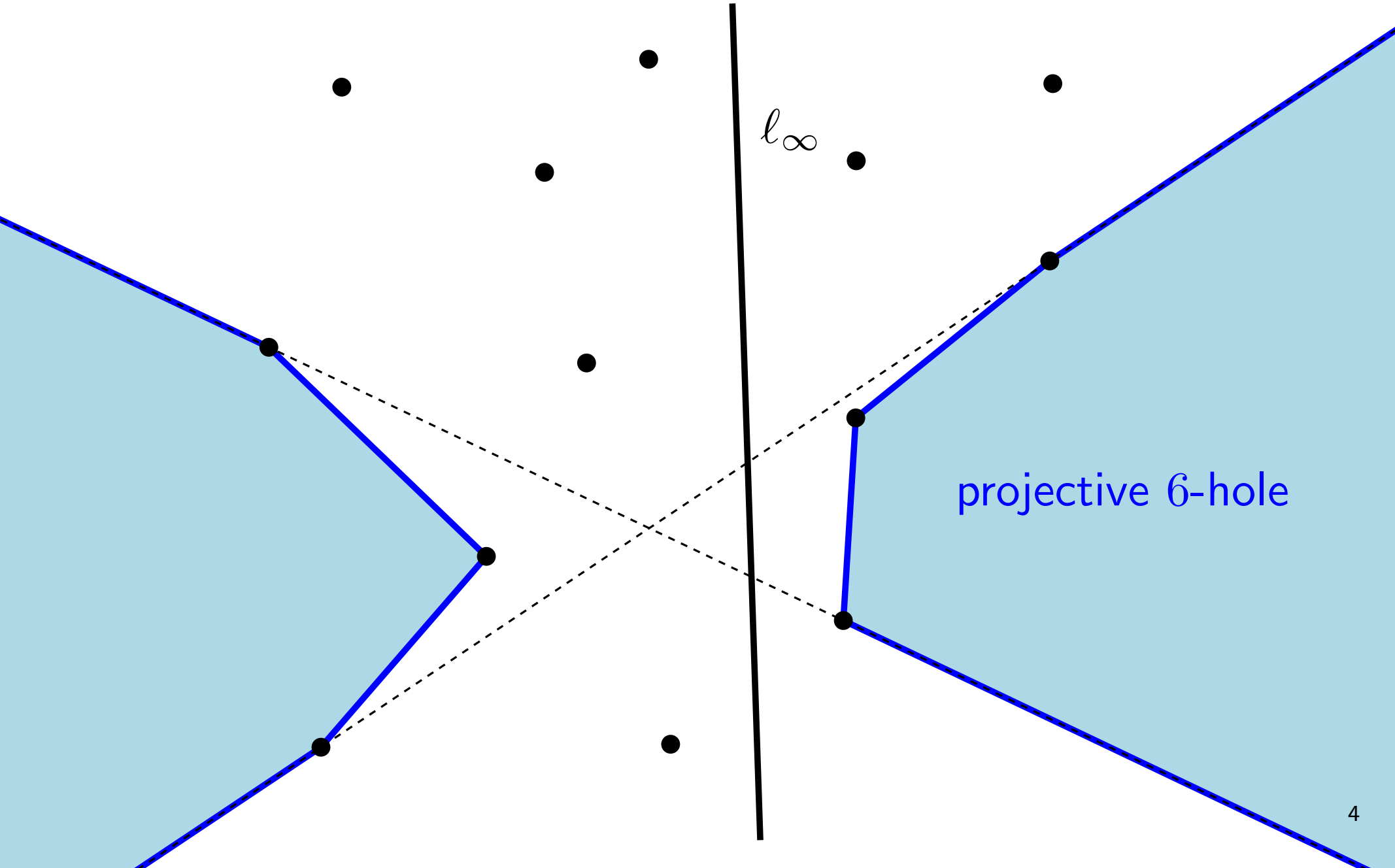

k-gons: Erdős–Szekeres Numbers

Thm (affine k-gons; Erdős and Szekeres '35; Holmsen, Mojarrad, Pach and Tardos '17):

$$2^{k-2} + 1 \leq g(k) \leq 2^{k+O(\sqrt{k \log k})}$$

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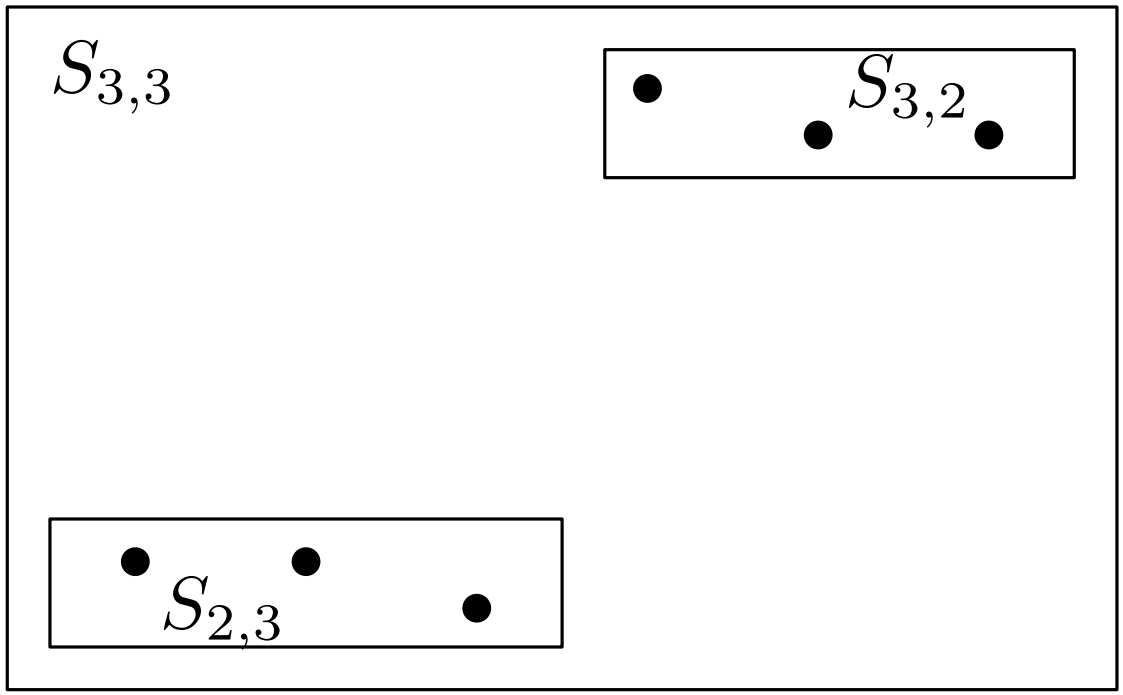
Proof: k-gons in Erdős–Szekeres construction

$S_{a,b}$ is defined recursively: $U = S_{a,b-1}$ and $L = S_{a-1,b}$ and has $\phi(a,b) = \binom{a+b-4}{a-2,b-2}$ points

$S_{a,b}$ has no a-cup and no b-cap



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if C is affine, statement follows directly from ES'35.
hence assume $C = C_1 \cup C_2$

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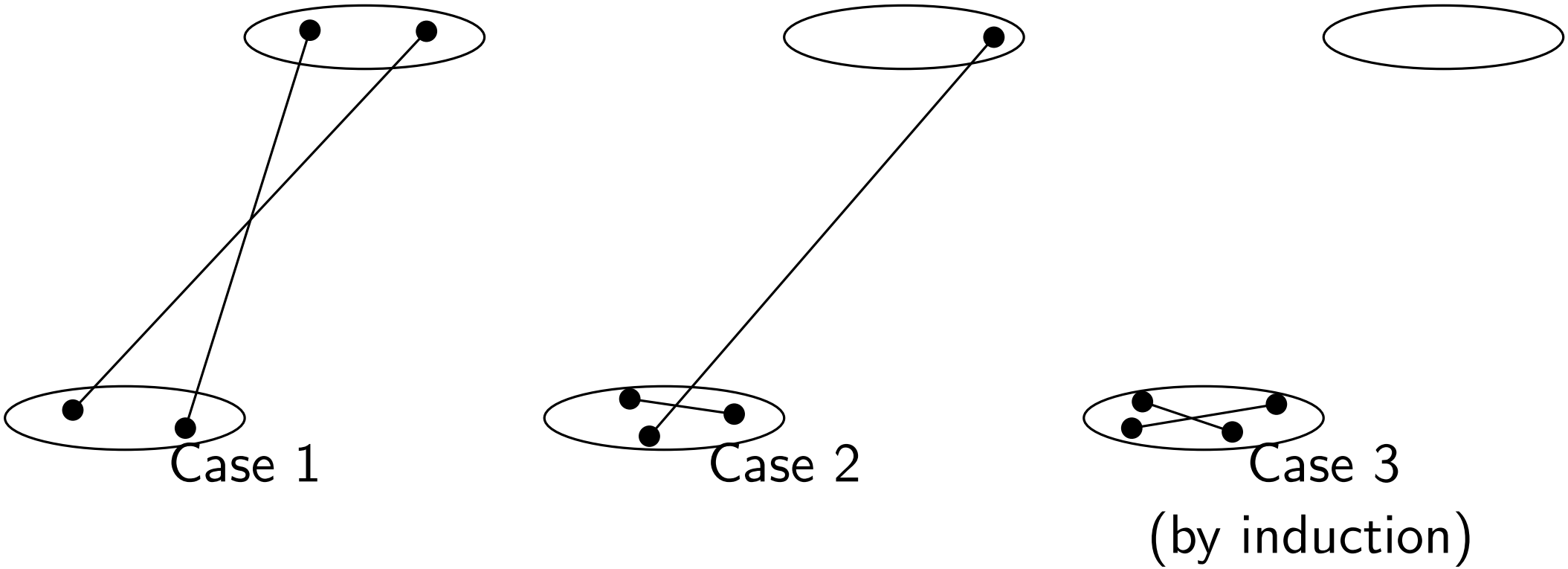

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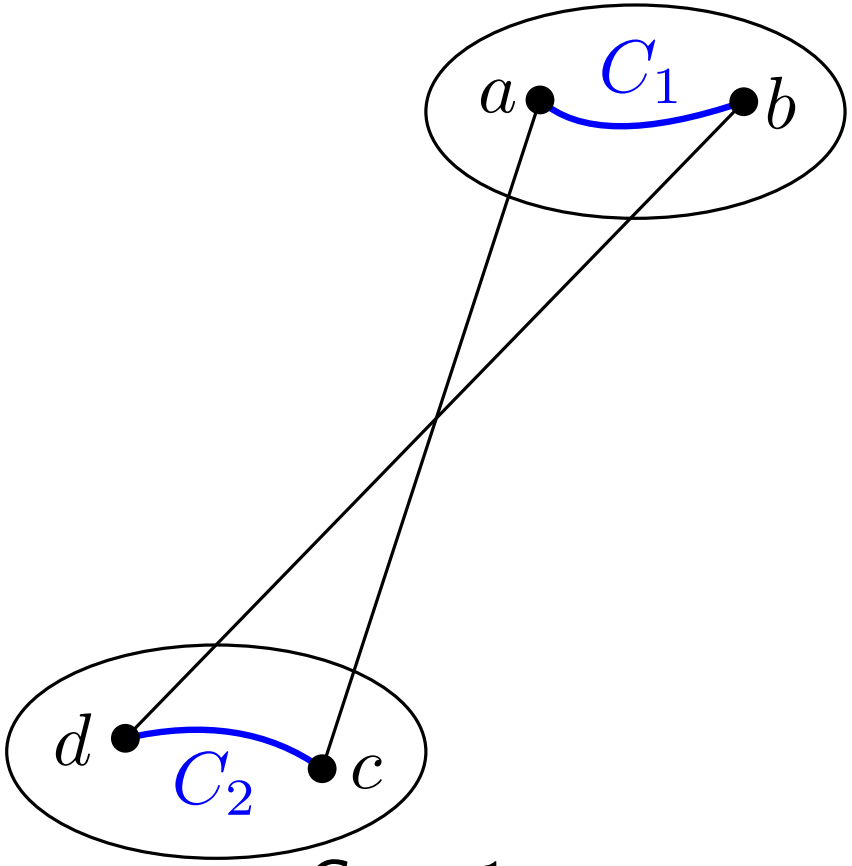

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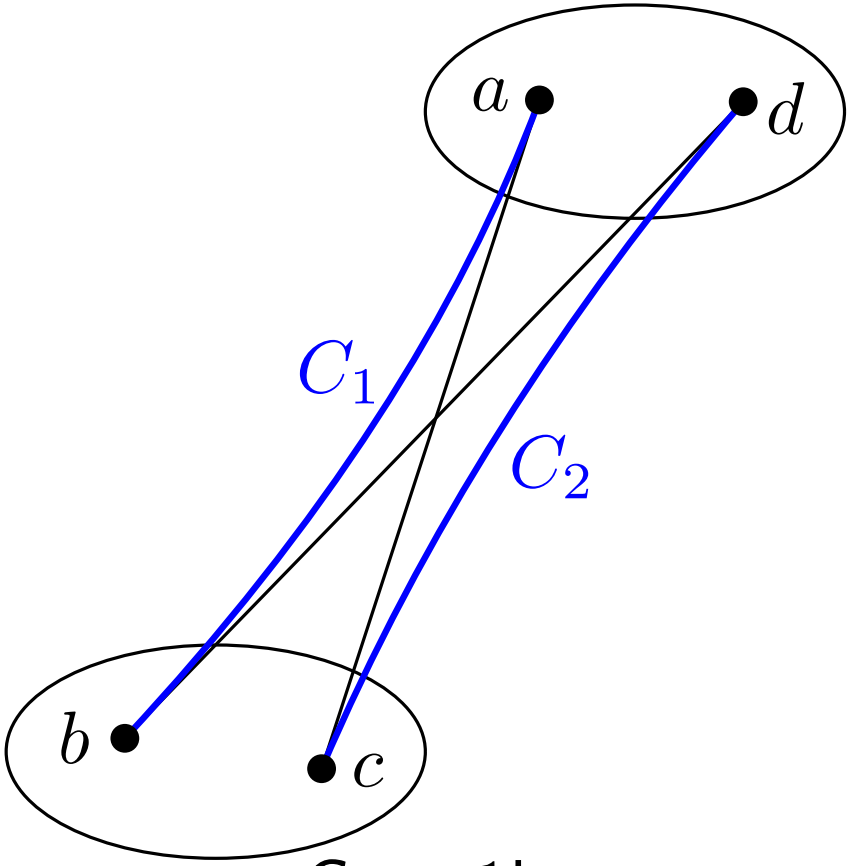

Case 1b

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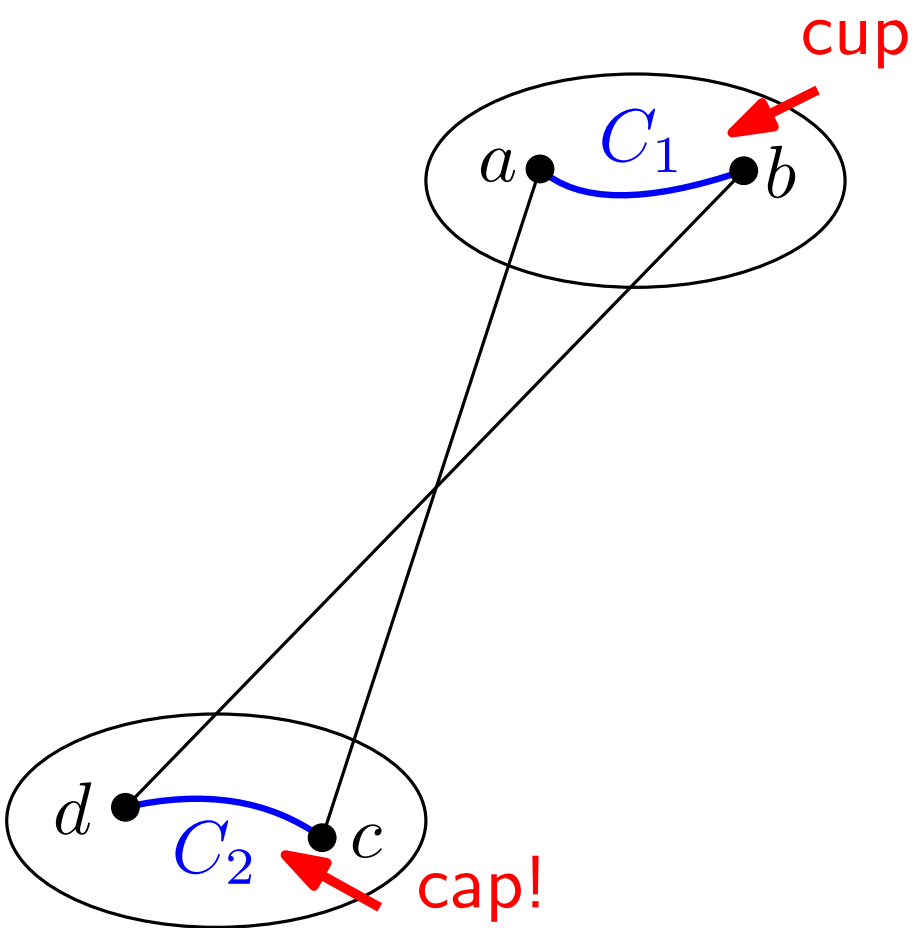

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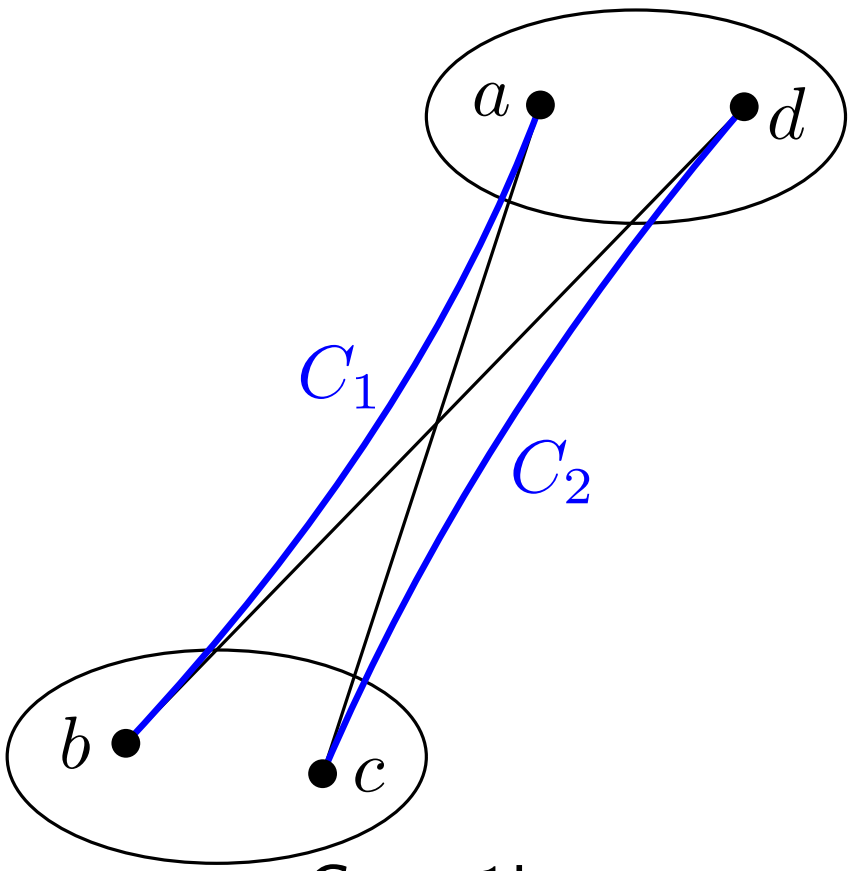

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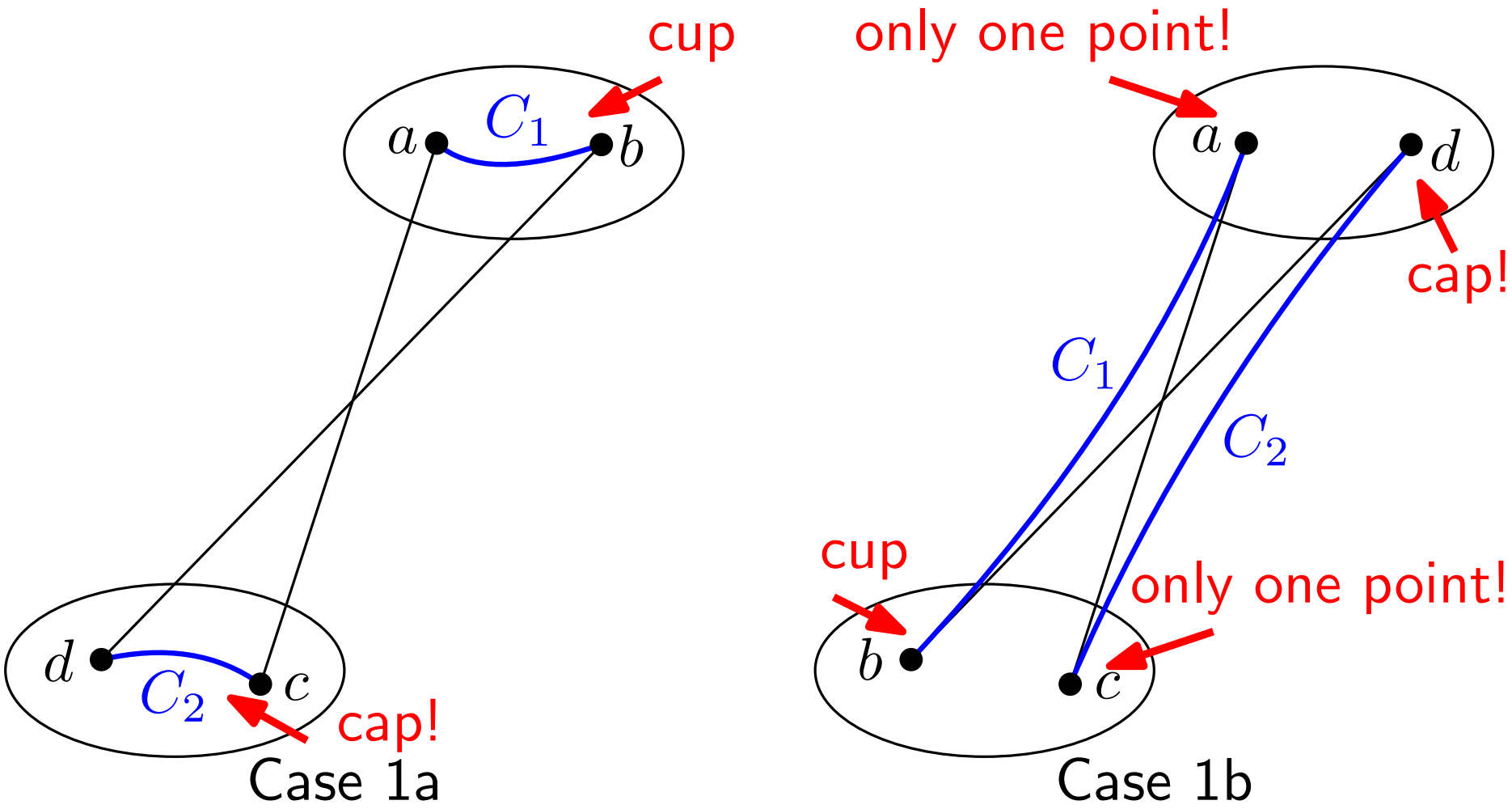

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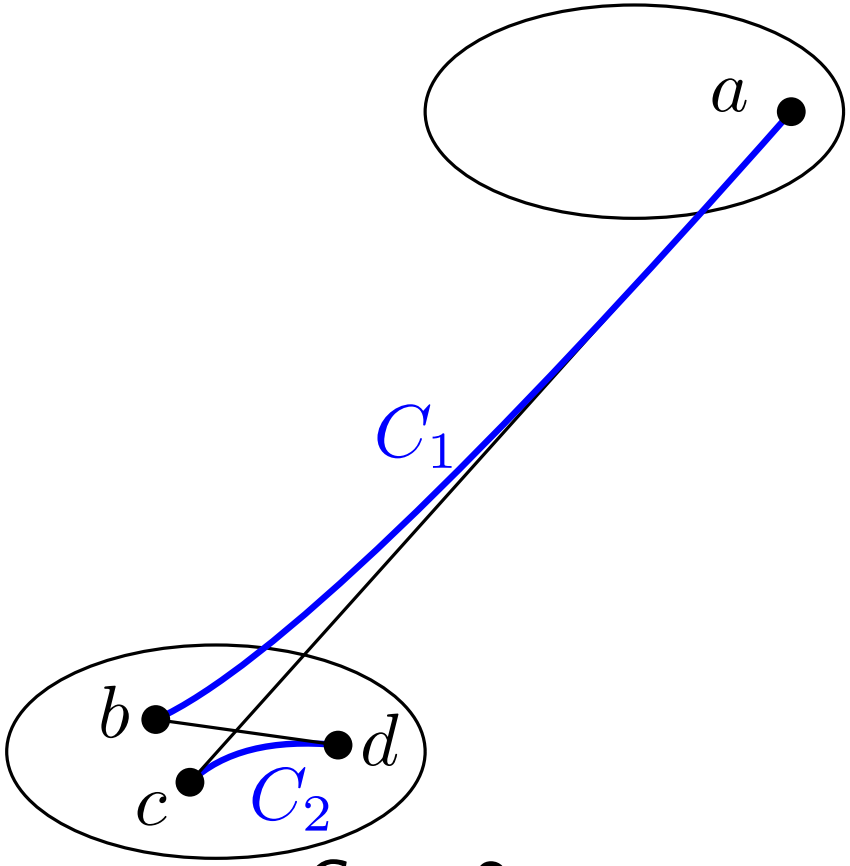

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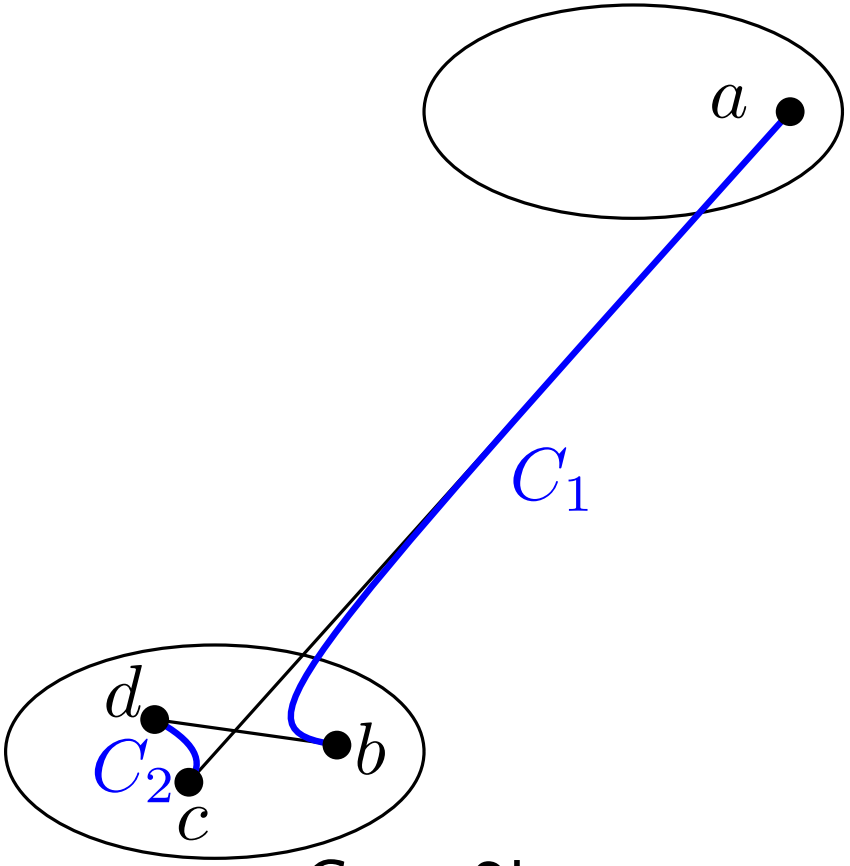

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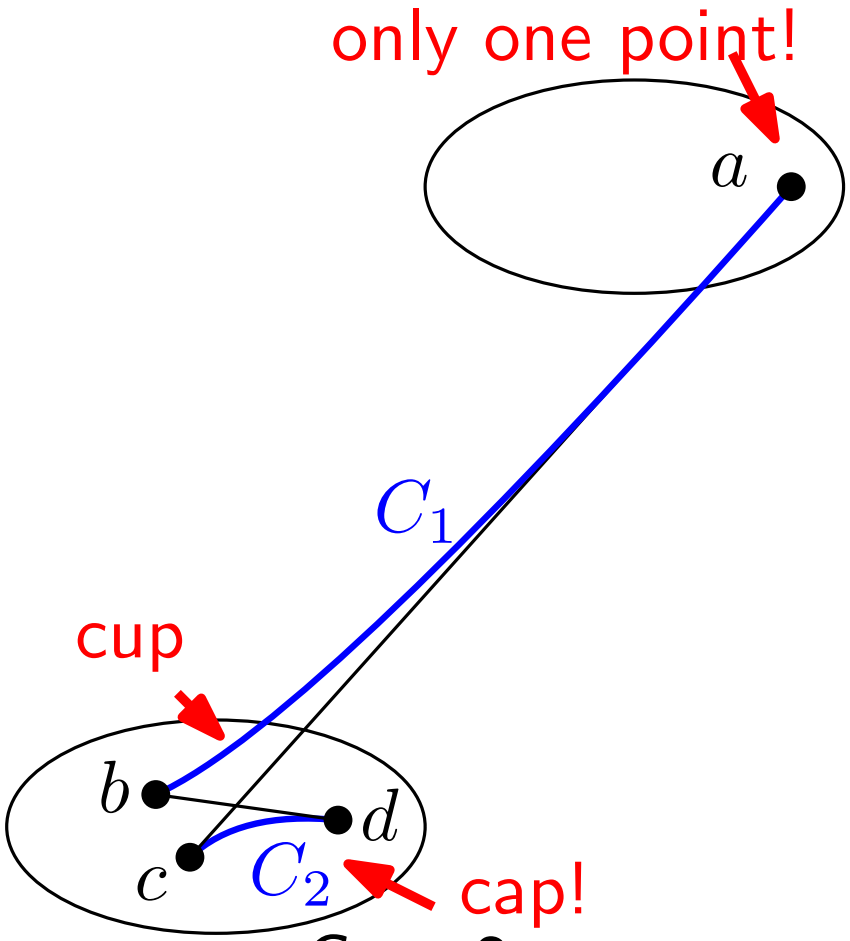

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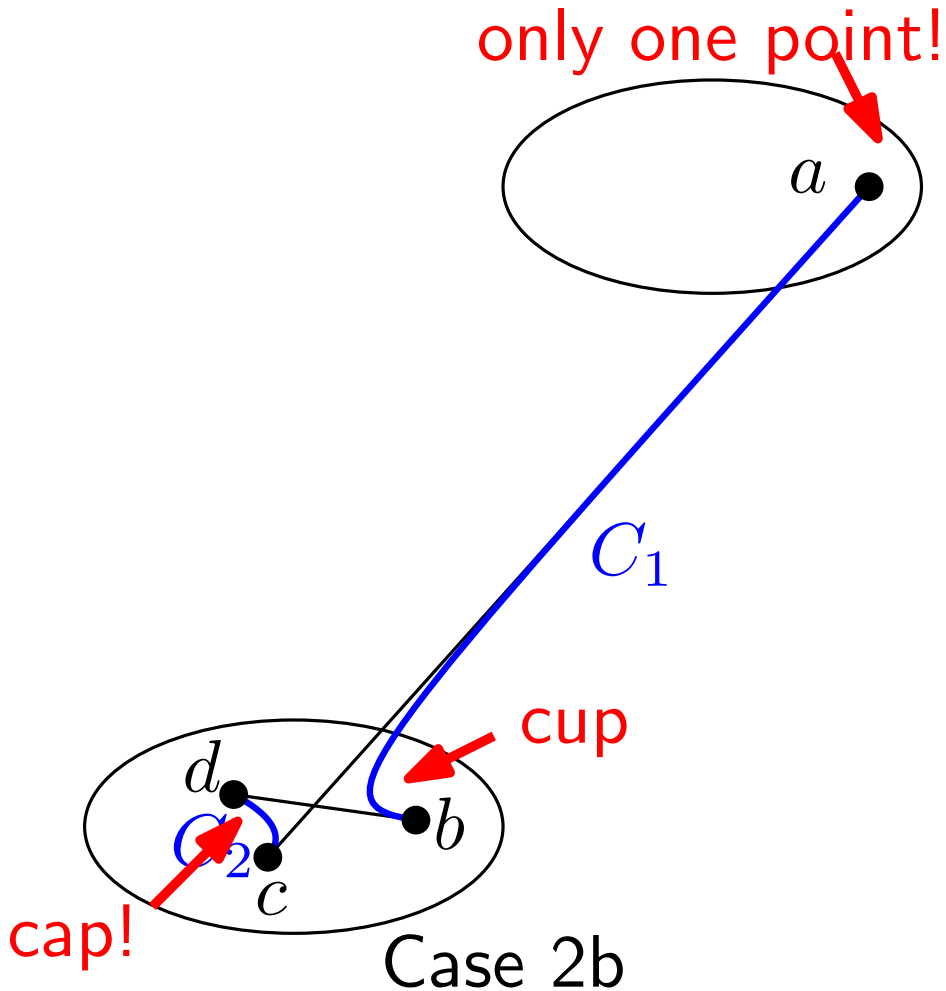

Case 2b

k-Holes in Horton Sets

Thm(affine, Horton '83, Bárány Füredi '87):
Let S be a (perfect) Horton set of size $n = 2^t$. Then

$$h_3(S) \leq 2n^2,\ h_4(S) \leq 3n^2,\ h_5(S) \leq 2n^2,\ h_6(S) \leq \frac{1}{2}n^2,$$

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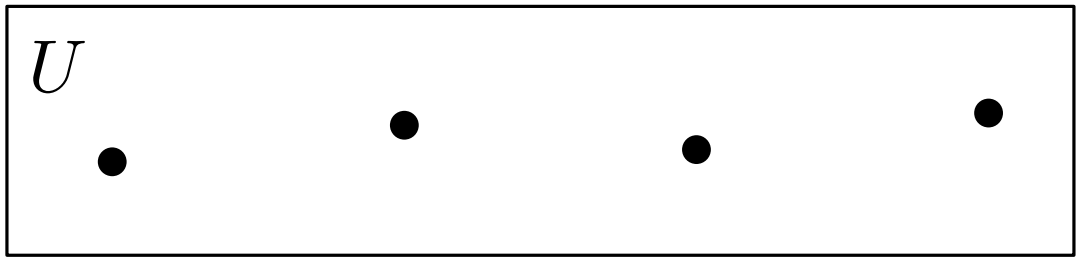
In particular, $h_3^p(S) = 4.25n^2 - O(n \log^2 n)$

Horton Sets and 3-Cups open from above

Horton set S_{t+1} consists of two copies L, U of S_t

left-to-right: points alternatingly in L and U

U high above L



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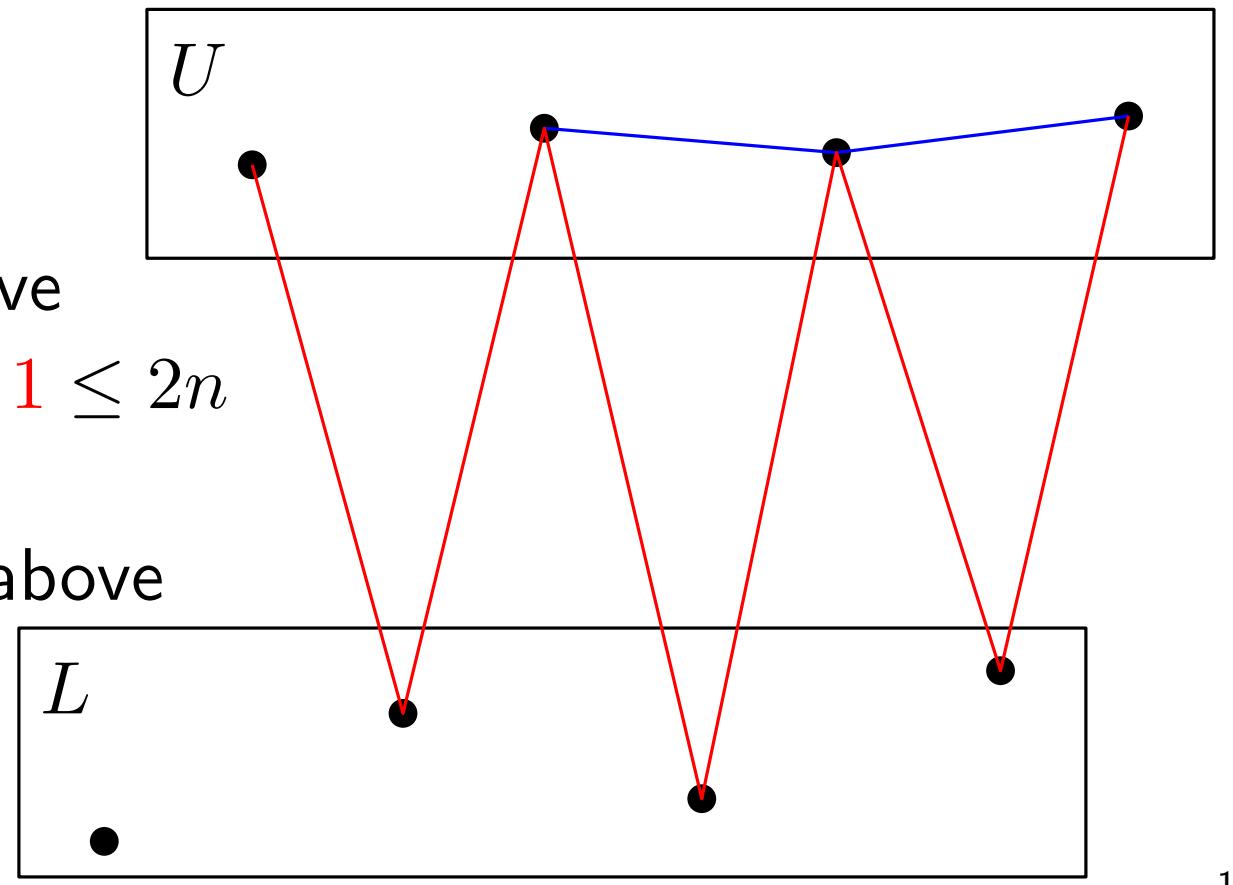


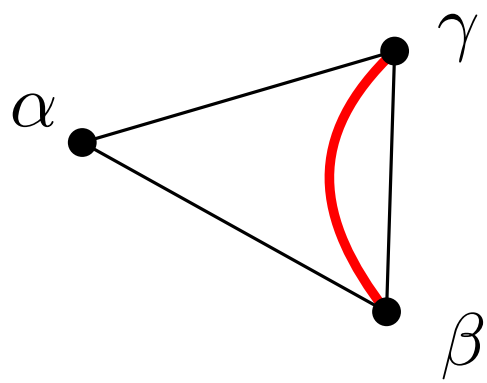
3-cups open from above

$U(n) = U(n/2) + n - 1 \leq 2n$

no 4-cups open from above

k-Holes in Horton Sets Proof

2 types of projective holes:



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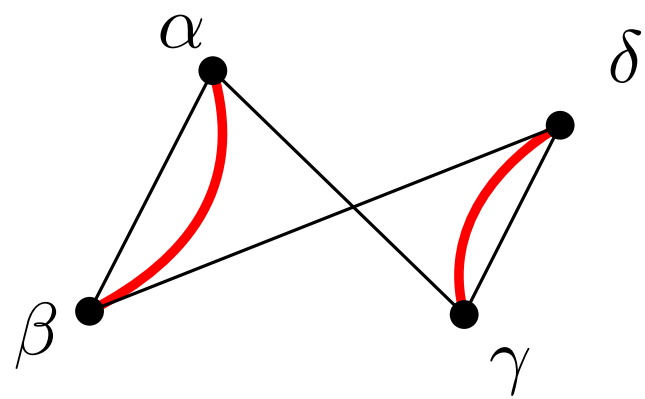

Case 2: 2+2 extremal

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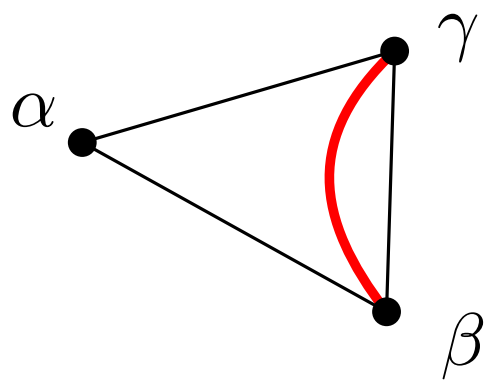

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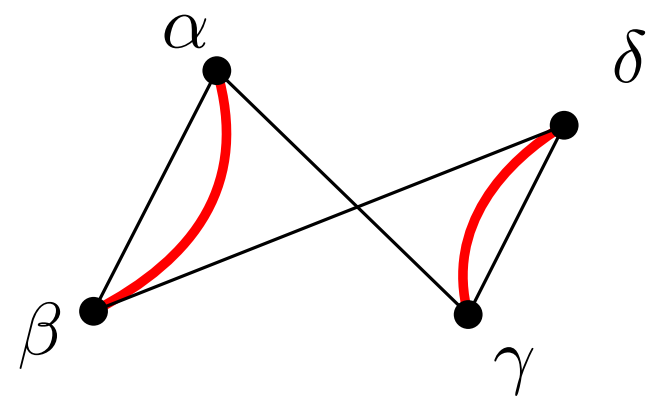

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points in L and points U, w.l.o.g. $\alpha \in U$

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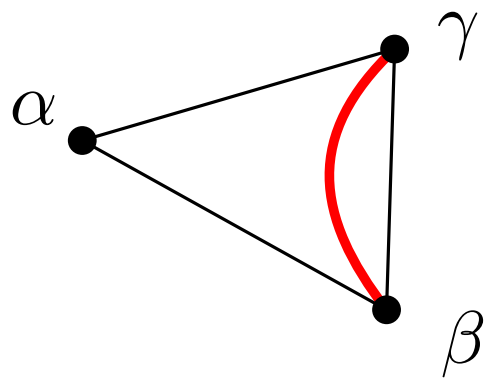

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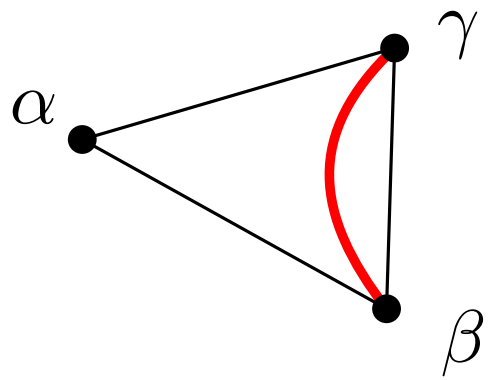

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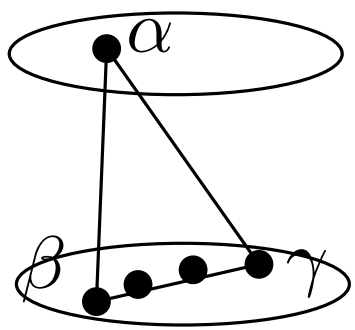

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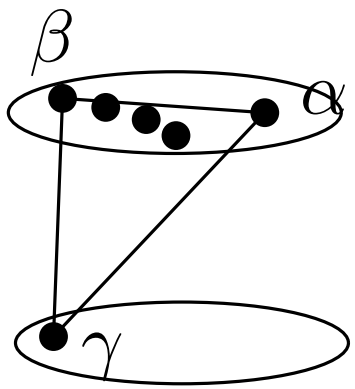

Case 1b

k-Holes in Horton Sets Proof

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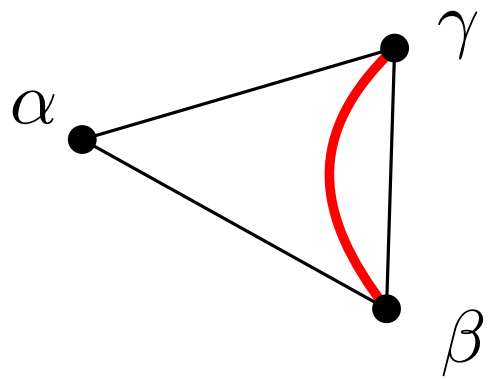

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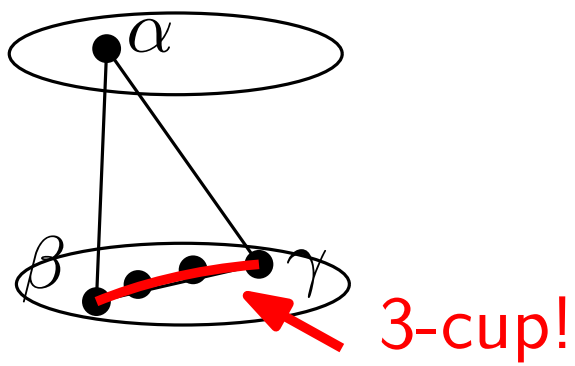
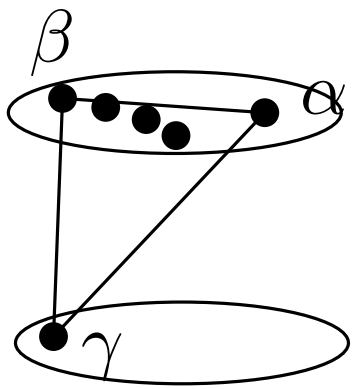


Case 1a: β, γ both in L $(k \leq 4)$



Case 1b

k-Holes in Horton Sets Proof

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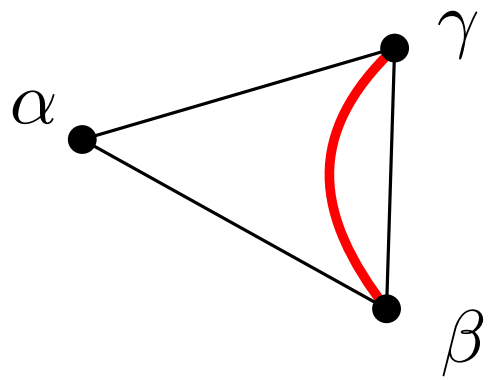

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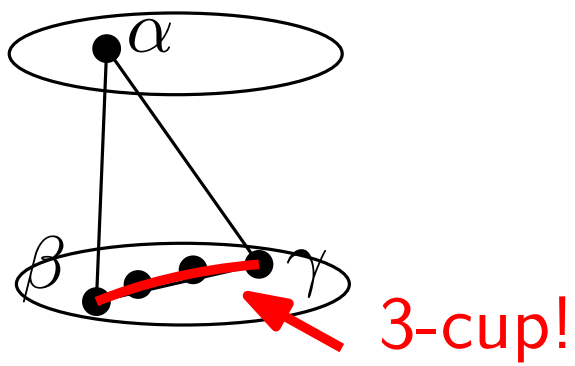

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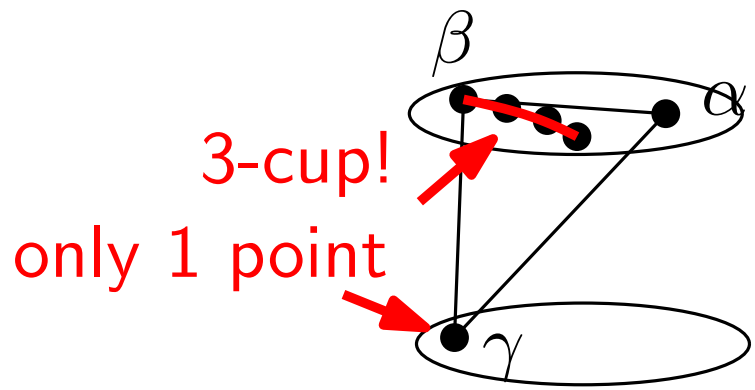

Case 1b $(k \leq 5)$

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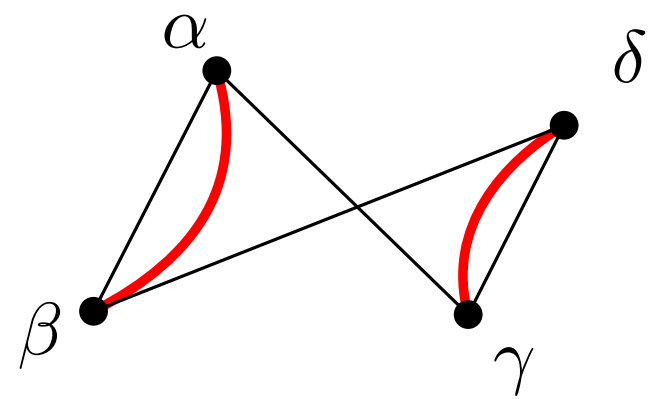

Case 2: 2+2 extremal

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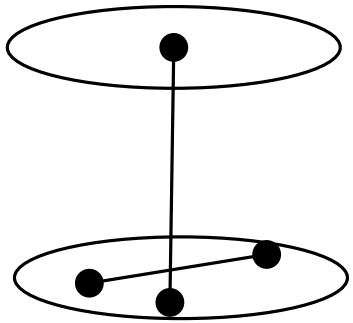
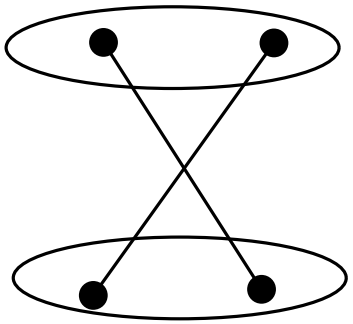


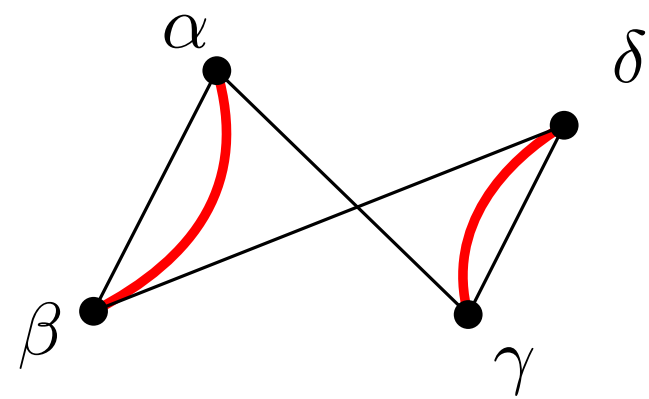
Case 2a



Case 2b

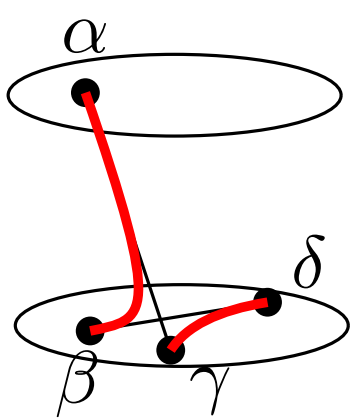
k-Holes in Horton Sets Proof

2 types of projective holes:

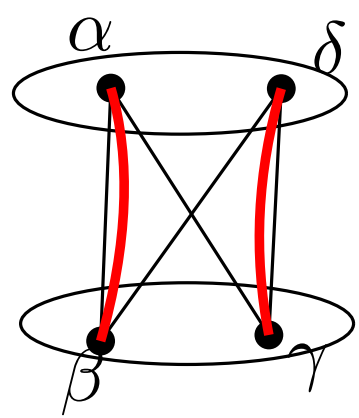


Case 2: 2+2 extremal

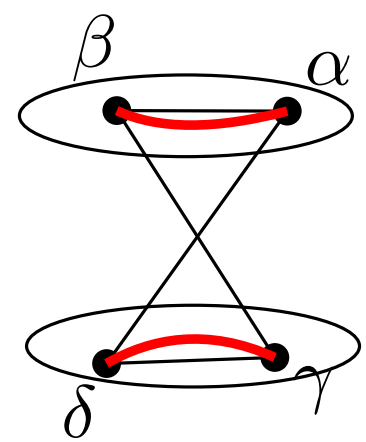
points in L and points U, w.l.o.g. $\alpha \in U$



Case 2a



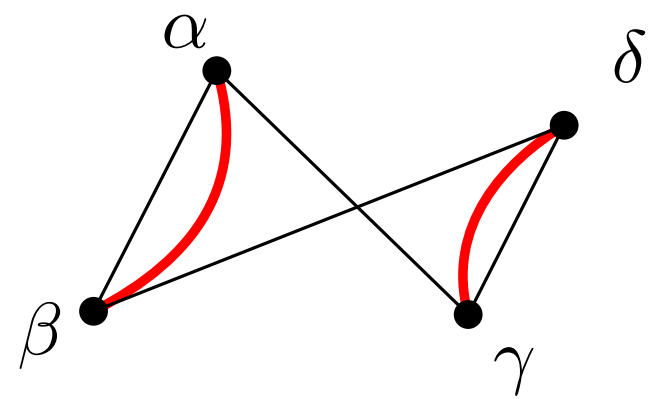
Case 2bα



Case 2bβ

k-Holes in Horton Sets Proof

2 types of projective holes:



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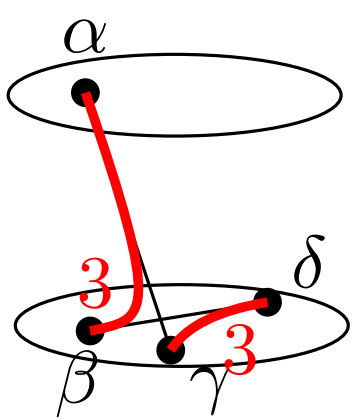
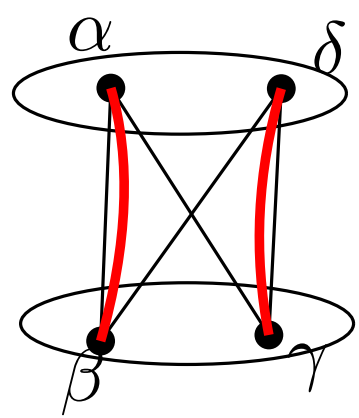
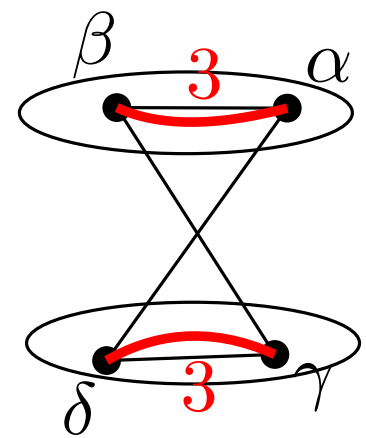


Case 2a $(k \leq 7)$



Case 2bα $(k = 4)$



Case 2bβ $(k \leq 6)$

k-Holes in General

$k = 3$: always exist

$k = 4$: $h_4(\geq 5) \geq 1 \Longrightarrow h_4^p(\geq 4) \geq 1$

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Harborth '78

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Harborth '78

weird behavior

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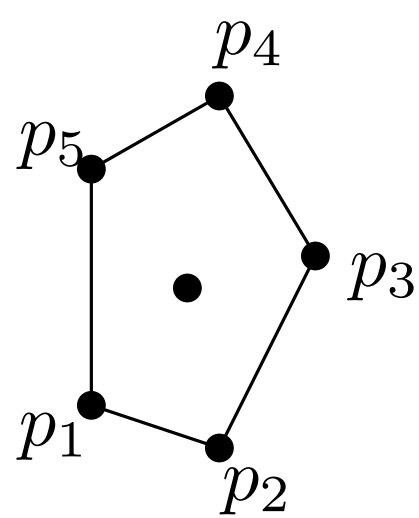
Proof: take projective 5-gon with minimal # inner points



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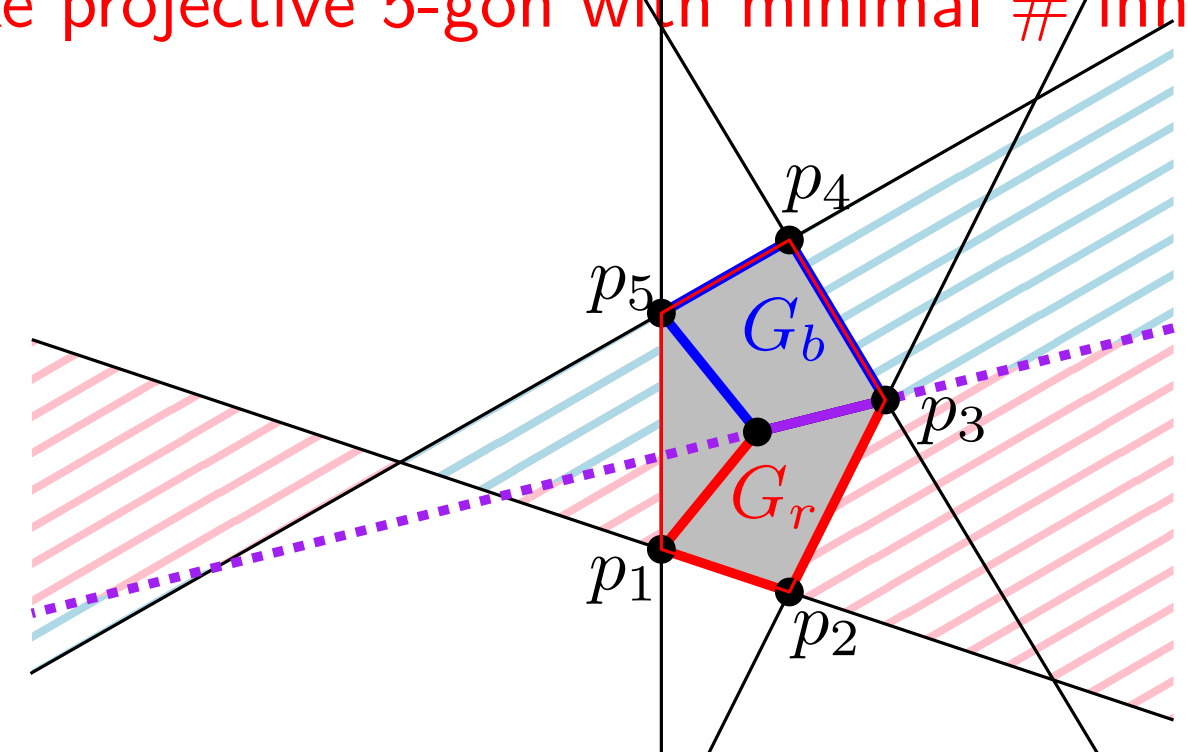

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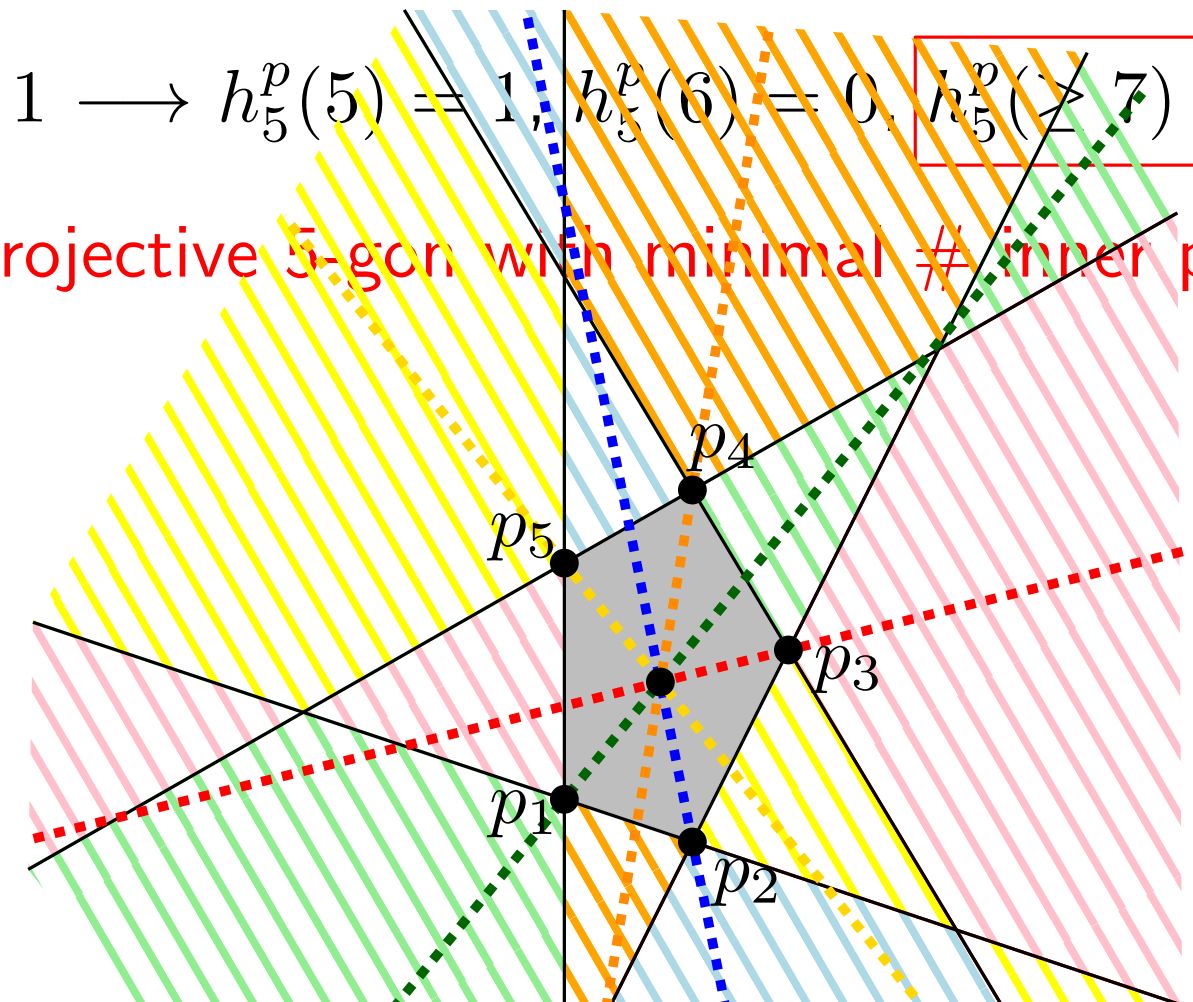

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$k = 6$:

Koshelev '09

$h_6(\geq \max\{g(9), 400\}) \geq 1$

$\longrightarrow h_6^p(\geq \max\{g^p(9), 400\}) \geq 1$

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$k \geq 8$: not exist (Horton sets)

$k = 7$: no affine, but projective 7-holes in Horton sets.

existence of projective 7-holes open problem!

k-holes: First and Second Moment Invariant

Thm (affine 1st moment, Edelman and Jamison '85):

$$\underbrace{h_0(P)}_{1} - \underbrace{h_1(P)}_{n} + \underbrace{h_2(P)}_{\binom{n}{2}} - h_3(P) \pm \ldots = \sum (-1)^k h_k(P) = 0$$

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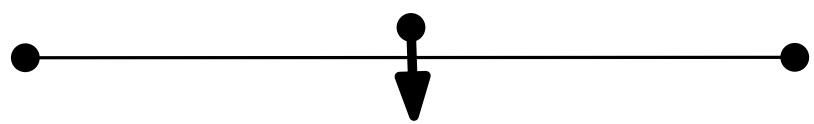
Proof idea: holds for convex position and invariant to flips



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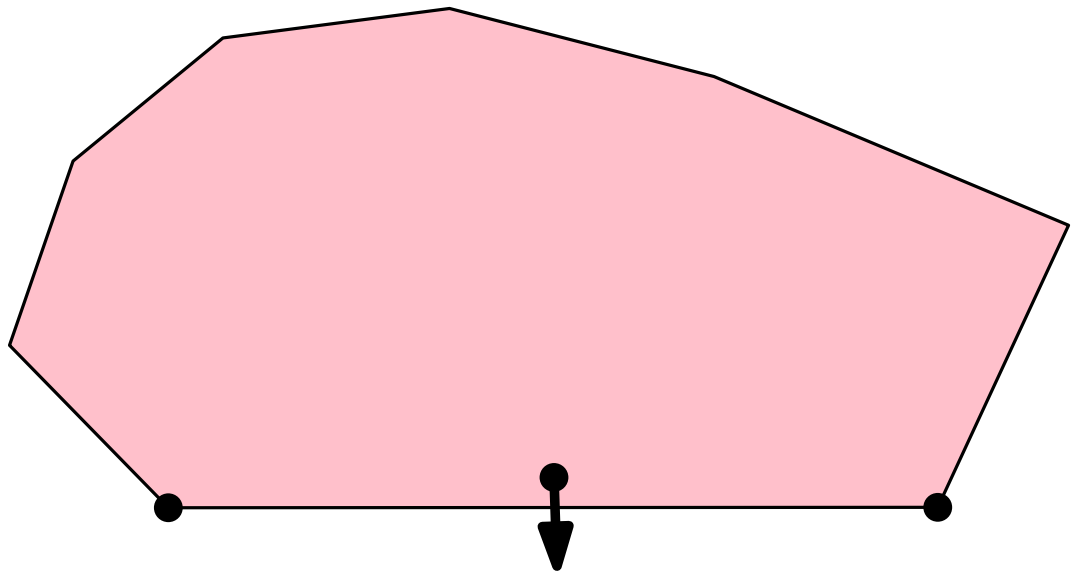

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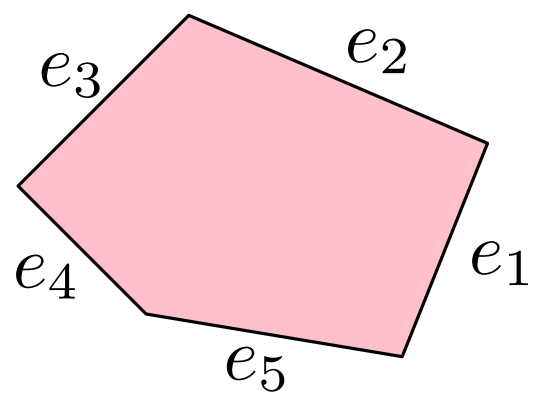
Idea: $k \cdot h_k(P) = \sum_e h_k(P; e)$



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Thm (projective, BSV'21+):

$$\sum (-1)^k h_k^p(P) = \alpha + \beta n + \gamma n^2$$

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Holes in Random Point Sets

In Affine:

$$EH_3 \sim 2n^2 \text{ [Valtr '95, Reitzner Temesvari '19]}$$

$$EH_k = \Theta(n^2) \text{ [BSV'19+'21]}$$

In Projective ('21+):

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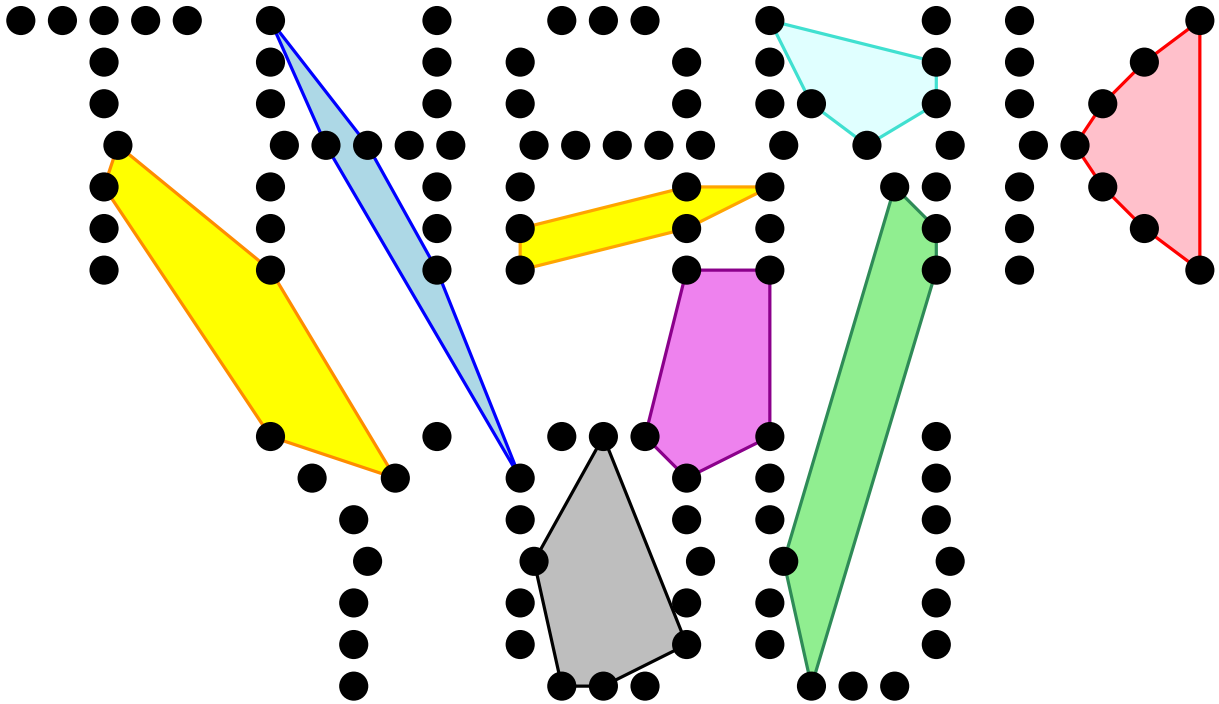
no proof for larger holes, but $\Theta(n^2)$ expected!

THANK YOU